

# Haokai Ding

Shenzhen, China

✉ ditang0125@gmail.com | 🏠 haokaiding.github.io | 📷 HaokaiDing

## Summary

Undergraduate researcher in robotics, focusing on underactuated grasping, idle-stroke and Peaucellier-based linkages, and contact-rich aerial manipulation. Building hardware prototypes and analytical models that bridge mechanism design with field-deployable manipulation. Incoming MSc in Robotics at MBZUAI.

## Education

### Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

M.SC. IN ROBOTICS (INCOMING)

Abu Dhabi, UAE

Fall 2026

### Tsinghua University | Shenzhen X-Institute

TSIEN EXCELLENCE IN ENGINEERING PROGRAM (TEEP) – X SCHOLAR

Shenzhen, China

Sep. 2023 – Present

- Joint training program in underactuated robotics and mechanism design.

### Shenzhen Technology University – Future Technology School

B.ENG. IN ELECTRONIC SCIENCE AND TECHNOLOGY

Shenzhen, China

Sep. 2022 – Jul. 2026

- GPA: 84.6/100. Research focus: LIBS spectroscopy and robotics systems.

## Research Experience

### State Key Lab of Mechanical Systems & Vibration, Shanghai Jiao Tong University

VISITING STUDENT – UAV PERCHING AND AERIAL MANIPULATION

Shanghai, China

Feb. 2026 – Present

- Advisor: Prof. Wei Dong.
- Working on UAV perching and contact-rich aerial manipulation.

### Shenzhen X-Institute, Tsinghua University

UNDERGRADUATE RESEARCHER – UNDERACTUATED ROBOTIC GRIPPERS

Shenzhen, China

2023 – Present

- Designed and tested underactuated grippers based on semi-Peaucellier linkages.
- Developed idle-stroke and delay-triggering mechanisms; validated prototypes via simulation and physical experiments.

### Shenzhen Technology University

UNDERGRADUATE RESEARCHER – LIBS SPECTROSCOPY FOR HEAVY-METAL ANALYSIS

Shenzhen, China

2022 – 2023

- Built and optimized a LIBS system for rapid heavy-metal adsorption analysis.
- Improved the calibration pipeline and acquisition accuracy.

## Publications

† Co-first author   \* Corresponding author

### SELECTED CONFERENCE PAPERS (ROBOTICS)

#### [P1] A Novel Gripper with Semi-Peaucellier Linkage and Idle-Stroke Mechanism for Linear Pinching and Self-Adaptive Grasping

**H. Ding**<sup>†</sup>, W. ZHANG<sup>\*</sup> | *IEEE/RSJ IROS 2025* | Nov. 2025

#### [P2] Semi-Peaucellier Linkage and Differential Mechanism for Linear Pinching and Self-Adaptive Grasping

**H. Ding**<sup>†</sup>, Z. CHEN, T. YANG<sup>\*</sup>, W. ZHANG<sup>\*</sup> | *IEEE CASE 2025* | Sep. 2025

#### [P3] An Idle-Stroke Underactuated Gripper with Independent Double-Parallelogram Linkages for Pinching and Adaptive Grasping

Y. CHEN, H. LUAN, **H. Ding**, W. ZHANG<sup>\*</sup> | *IEEE RAAI 2025* | Mar. 2026

#### [P4] BioCrest – A Flexible Adaptive Robotic Hand Based on Idle-Stroke Mechanism

**H. Ding**<sup>†</sup>, X. ZHONG, W. HUANG, T. YANG<sup>\*</sup>, W. ZHANG<sup>\*</sup> | *IEEE ROBIO 2025* | Feb. 2026

## [P5] GLINT: An Idle-Stroke Grasp-and-Lift Hand for In-Hand Manipulation

**H. DING**, H. LUAN, T. YANG\*, W. ZHANG\* | *IEEE ICCMA 2025* | Feb. 2026

## [P6] Peaucellier Gripper: A Novel Underactuated Gripper for Linear Pinching and Self-Adaptive Grasp

**H. DING**<sup>†</sup>, J. FAN, Z. ZHU, Y. ZHAO, K. CHEN\*, W. ZHANG\* | *IEEE ICARM 2025* | Dec. 2025

## [P7] An Underactuated Gripper with Grasshopper-Inspired Linkages and Delay Triggering for Pinching and Adaptive Grasping

**H. DING**<sup>†</sup>, Y. CHEN<sup>†</sup>, D. JIA, W. ZHANG\* | *IEEE ICIA 2025* | Dec. 2025

## OTHER PUBLICATIONS

## [P8] Recent Progress on the Research of 3D Printing in Aqueous Zinc-Ion Batteries

Y. LIU<sup>†</sup>, **H. DING**<sup>†</sup>, H. CHEN, H. GAO, J. YU, F. MO, N. WANG\* | *Polymers (MDPI)* 17(15):2136 | Aug. 2025

## [P9] Rapid Analysis of Heavy-Metal Element Adsorption by SCG Based on LIBS Technology

H. XIE, L. YOU, X. FANG, L. LI, **H. DING**, Z. ZHOU, G. ZHANG, D. ZHANG | *E3S Web of Conferences*, 615 (2025) 02010 | Feb. 2025

## Honors & Awards

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|------|-----------------------------------------------------------------------------------------------------|------------------|
| 2025 | <b>Undergraduate Champion</b> , ASME Student Mechanism & Robot Design Competition                   | USA              |
| 2025 | <b>National Scholarship</b> , Ministry of Education of China                                        | China            |
| 2024 | <b>President's Award</b> , Shenzhen Technology University                                           | Shenzhen, China  |
| 2024 | <b>Research and Innovation Award (First Prize)</b> , Shenzhen Technology University                 | Shenzhen, China  |
| 2023 | <b>Gold Prize</b> , 9th China "Internet+" Innovation Competition (Guangdong Division)               | Guangdong, China |
| 2023 | <b>Second Prize</b> , 17th "Challenge Cup" Extracurricular Science Competition (Guangdong Division) | Guangdong, China |
| 2023 | <b>Research and Innovation Award (First Prize)</b> , Shenzhen Technology University                 | Shenzhen, China  |

## Professional Service

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|------|---------------------------------------------------------------------------------------------------------|-----------------|
| 2025 | <b>Conference Reviewer</b> , IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) |                 |
| 2025 | <b>Session Chair</b> , IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)       | Hangzhou, China |
| 2025 | <b>Conference Reviewer</b> , IEEE International Conference on Automation Science and Engineering (CASE) |                 |